Calculus I

Computation problems #4

Topics:

- Rate of change
- Derivatives of trigonometrical functions
- The Chain Rule
- Implicit differentiation
- Related rates
- Linearization

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Problem 1. Find $dr/d\theta$ for the following functions:
a) $r = 4 - \theta^2 \sin\theta$; b) $r = \sec\theta \csc\theta$; c) $r = \dfrac{\sin\theta + \cos\theta}{\cos\theta}$.

Problem 2. Is there a value of b that makes

$$g(x) = \begin{cases} x + b, & x < 0, \\ \cos x, & x \geq 0, \end{cases}$$

continuous at $x = 0$? differentiable at $x = 0$? Justify your answers.

Problem 3. Find the derivatives of the following functions:

a) $p(t) = \sqrt{3 - t}$.

b) $y = x^2 \sin^4 x + x \cos^{-2} x$.

c) $f(\theta) = \left(\frac{\sin\theta}{1+\cos\theta}\right)^2$.

d) $q(t) = \sin\left(\frac{t}{\sqrt{t+1}}\right)$.

Problem 4. Find dy/dt of $y = \left(1 + \tan^4\left(\frac{t}{12}\right)\right)^3$.

Problem 5. Find y'' of $y = \left(1 + \frac{1}{x}\right)^3$.

Problem 6. Use implicit differentiation to find dy/dx.

a) $x^2y + xy^2 = 6$.

b) $x^2(x-y)^2 = x^2 - y^2$.

c) $x + \tan(xy) = 0$.

Problem 7. *(Volume)* The radius r and height h of a right circular cylinder are related to the cylinder's volume V by the formula $V = \pi r^2 h$.

a) How is dV/dt related to dh/dt if r is constant?

b) How is dV/dt related to dr/dt if h is constant?

c) How is dV/dt related to dr/dt and dh/dt if neither r or h is constant.

Problem 8. *(Distance)* Let x and y be differentiable functions of t and let $s = \sqrt{x^2 + y^2}$ be the distance between the points $(x, 0)$ and $(0, y)$ in the xy-plane.

a) How is ds/dt related to dx/dt if y is constant?

b) How is ds/dt related to dx/dt and dy/dt if neither x nor y is constant?

c) How is dx/dt related to dy/dt if s is constant?

Problem 9. *(Area)* the area A of a triangle with sides of lengths a and b enclosing an angle of measure θ is

$$A = \frac{1}{2}ab\sin\theta.$$

a) How is dA/dt related to $d\theta/dt$ if a and b are constant?

b) How is dA/dt related to $d\theta/dt$ and da/dt if only b is constant?

c) How is dA/dt related to $d\theta/dt$, da/dt and db/dt if none of a, b, and θ are constant?

Problem 10. Show that the linearization of $f(x) = (1+x)^k$ at $x = 0$ is $L(x) = 1 + kx$.

Problem 11. *(Faster than a calculator)* Use the approximation $(1 + x)^k \approx 1 + kx$ to estimate the following.

a) $(1.0002)^{50}$

b) $\sqrt[3]{1.009}$.

Problem 12. *(Estimating volume)* Estimate the volume of material in a cylindrical shell with height 30 in, radius 6 in, and shell thickness 0.5 in.

Problem 13. Is the derivative of

$$h(x) = \begin{cases} x^2 \sin(1/x), & x \neq 0, \\ 0, & x = 0, \end{cases}$$

continuous at $x = 0$? How about the derivative of $k(x) = xh(x)$? Justify your answer.

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Answer to selected problems

2. $b = 1$ and $\nexists\, b$

11. (a) 1.01 (b) 1.003.

13. No; Yes.